

Datasci 261



Machine Learning at Scale

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This course will provide an accessible introduction to MapReduce frameworks such as Hadoop and Spark and their potential to revolutionize academic and commercial data science practices through scale. Conceptually, the course includes two simultaneous components. The first covers fundamental concepts of MapReduce parallel computing via Hadoop and Spark. The second focuses on hands-on algorithmic design and development in parallel computing environments such as Spark; developing algorithms from scratch, such as decision-tree learning; graph-processing algorithms such as PageRank and shortest path; gradient descent algorithms such as linear regression and classification, linear support vector machines; matrix factorization, and unsupervised machine learning algorithms. Industrial applications and deployments of MapReduce parallel compute frameworks from various fields, including advertising, finance, healthcare, and search engines, help tie these components together.

[Click here for sample syllabus.](https://learn.berkeley.edu/courses/395/files/16580?wrap=1) (<https://learn.berkeley.edu/courses/395/files/16580?wrap=1>). [↓](https://learn.berkeley.edu/courses/395/files/16580/download?download_frd=1)
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Module	Videos
1 - Introduction and Motivation for Machine Learning at Scale	 (https://learn.berkeley.edu/courses/395/pages/datasci261-dot-1)
2 - Parallel Computation Frameworks	 (https://learn.berkeley.edu/courses/395/pages/datasci261-dot-2)

Module	Videos
	dot-2)
3 - Map-Reduce Algorithm Design	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-3)
4 - Introduction to Spark with RDDs: Part 1	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-4)
5 - Introduction to Spark with RDDs: Part 2	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-5)
6 - Distributed Supervised Machine Learning: Part I	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-6)
7 - Distributed Supervised Machine Learning: Part II	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-7)
8 - Big Data Systems and Pipelines	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-8)
9 - Graph Algorithms at Scale	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-9)
10 - Large Scale Graph Processing: Random Walks, Page Rank	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-10)
11 - Alternating Least Squares, Various Optimizations in Spark and Working with Spark MLlib	 .https://learn.berkeley.edu/courses/395/pages/datasci261-dot-11)

Module	Videos
12 - Distributed Decision Trees for Classification and Regression	 https://learn.berkeley.edu/courses/395/pages/datasci261-dot-12
13 - Streaming Systems	 https://learn.berkeley.edu/courses/395/pages/datasci261-dot-13